

# F1 Educational Carnival & Experience

## Innovation Statement

F1 Educational Carnival & Experience rethinks Formula 1 learning as a spatial, browser-native experience rather than a collection of pages, videos, or isolated data views. It addresses a common limitation of conventional sports information websites: history, engineering concepts, data, and activities are usually consumed separately and without awareness of the visitor's context. Here, exploration itself becomes the information architecture. Users navigate a large 3D Carnival and enter a connected Exhibition Hall in first person, moving through tyre, chassis, track, franchise-history, championship, quiz, gaming, and historic-car experiences as places with purpose.

The central innovation is a contextual guidance model that connects where a visitor is, what they are looking at, and how information is delivered. A spatial location engine identifies Carnival zones and interaction points, including the exhibition entrance, educational areas, championship and examination sections, game area, map, AI point, and exhibition cars. These triggers can connect position to visual interfaces and relevant narration. In the Exhibition Hall, proximity to historic cars can activate associated prerecorded audio, creating a continuous relationship between 3D position, exhibit context, visual information, and sound rather than requiring users to search a separate interface.

AI extends this model as an in-world F1 guide, not a generic chatbot. Google Gemini is accessed through a server-side Next.js route and is bounded by an F1 domain guard, spatial/location-aware query handling, grounded contextual information, and a local F1 knowledge-base fallback. Questions outside Formula 1 are deliberately declined with an explanation of the assistant's scope. The guide is embodied in two complementary interfaces: an AI terminal inside the Exhibition Hall and a 3D Smart Guide phone carried in the first-person view. The phone stays lowered during exploration and is raised when activated, making assistance feel like a situated tool within the environment. Text-based interaction is complemented by text-to-speech output.

The project combines learning and participation: technical and historical exhibits, a 2000–2025 champions section, educational boards leading to a 40-question quiz, historic vehicles, and a small F1

racing game. This combination lets visitors shift between discovery, explanation, assessment, and play without leaving the same navigable world.

Its technical significance lies in delivering this integrated experience on the web. Built with Next.js, React, TypeScript, Three.js and React Three Fiber over WebGL/WebGL2, it uses GLB/GLTF assets with Draco compression, dynamic CanvasTextures for in-world screens, custom kinematic first-person movement with AABB collision detection, and pointer-lock navigation. Web Speech and Web Audio APIs support narration, background sound, ducking, hall reverb, filtering, and dynamic compression. A GraphicsManager evaluates browser-accessible GPU information, CPU thread count, and device memory to adjust pixel ratio, shadows, draw and fog distances, and rendering quality. Together, these choices make a large interactive 3D learning environment deployable as a hosted browser experience while adapting its visual cost to available hardware.